

Downward density in the hyperenumeration degrees

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Table of contents

- 1 Enumeration reducibility
- 2 Downwards density in $\mathcal{D}_e$
- 3 Computability in $L_{\omega_1^{CK}}$
- 4 Hyperenumeration reducibility
- 5 Downwards density in $\mathcal{D}_{he}$
- 6 Other reducibilities

Table of contents

- 1 Enumeration reducibility
- 2 Downwards density in $\mathcal{D}_e$
- 3 Computability in $L_{\omega_1^{CK}}$
- 4 Hyperenumeration reducibility
- 5 Downwards density in $\mathcal{D}_{he}$
- 6 Other reducibilities

Enumeration reducibility

Definition (Friedberg and Rogers 1959)

A set A is *enumeration reducible* to a set B ($A \leq_e B$), if there a program that transforms any enumeration of B into an enumeration of A .

In practice, we use that $A \leq_e B$ if and only if there is a c.e. set of axioms W such that

$$x \in A \iff \exists \langle x, u \rangle \in W [D_u \subseteq B]$$

where $(D_u)_{u \in \omega}$ is a listing of all finite sets by strong indices.

Example

Consider the halting set $K = \{\langle e, n \rangle : \varphi_e(n) \downarrow\}$.

$K \leq_e A$ for any A since K is c.e.

$\overline{K} \not\leq_e K$ since $\overline{K}$ is not c.e.

Degree structure

- Like with Turing reducibility, $\leq_T$, we have that $\leq_e$ is a pre-order and taking equivalences classes give us a degree structure $\mathcal{D}_e$.
- The lowest element of $\mathcal{D}_e$ is 0_e which comprises of the c.e. sets.
- From an effective listing of c.e. sets $(W_e)_{e \in \omega}$, we obtain an effective listing of enumeration operators $(\Psi_e)_{e \in \omega}$, defined by $A = \Psi_e(B)$ if $A \leq_e B$ via the set of axioms W_e .
- $\Psi_e(A)$ is always a set.
- Ψ_e is monotonic: if $B \subseteq A$ then $\Psi_e(B) \subseteq \Psi_e(A)$.

Relationship with Turing reducibility

Enumeration reducibility has a close relationship with Turing reducibility:

- For any sets A, B we have that A is c.e. in B if and only if $A \leq_e B \oplus \overline{B}$.
- For any sets A, B we have that $A \leq_T B$ if and only if $A \oplus \overline{A} \leq_e B \oplus \overline{B}$.
- The map $A \mapsto A \oplus \overline{A}$ induces an embedding of $\mathcal{D}_T$ in $\mathcal{D}_e$.
- The image of this embedding is known as the *total* degrees, TOT , and is a proper subclass of the enumeration degrees.

Theorem (Selman's theorem)

For any sets A, B we have that

$$A \leq_e B \iff \{X : A \leq_e X \oplus \overline{X}\} \supseteq \{X : B \leq_e X \oplus \overline{X}\}$$

Table of contents

- 1 Enumeration reducibility
- 2 Downwards density in $\mathcal{D}_e$
- 3 Computability in $L_{\omega_1^{CK}}$
- 4 Hyperenumeration reducibility
- 5 Downwards density in $\mathcal{D}_{he}$
- 6 Other reducibilities

Downwards density

Theorem (Gutteridge '71)

For any enumeration degree $a > 0_e$ there is an enumeration degree b such that $0_e < b < a$

Proof.

Lemma (Gutteridge operator)

There is an enumeration operator, Θ , such that for any set X :

- If $X \leq_e \Theta(X)$ then X is c.e.*
- $X \leq_e \Theta(X) \oplus \emptyset'$.*

Lemma (Δ_2^0 case)

If A is Δ_2^0 and not c.e, then there is an enumeration operator Φ such that $\emptyset <_e \Phi(A) <_e A$.

Lemma (Gutteridge operator)

There is an enumeration operator, Θ , such that for any set X :

- *If $X \leq_e \Theta(X)$ then X is c.e.*
- *$X \leq_e \Theta(X) \oplus \overline{\emptyset}'$.*

Proof.

Build c.e. set $B \subseteq \omega$ defined by $B = \bigoplus_k \{x < n_k\}$.

$\Theta(X) = B \cup \{\langle k, n_k \rangle : k \in X\}$.

$k \in X \iff \exists n[\langle k, n \rangle \in \overline{B} \cap \Theta(X)]$, so $X \leq \Theta(X) \oplus \emptyset'$.

For all $m > e$ and finite $D \subseteq m$ we have the following requirement:

$$\mathcal{R}_{e,m,D} : m \in \Psi_e(\Theta(D)) \iff m \in \Psi_e(\Theta(D \cup \{n : n \geq m\}))$$



Lemma (Δ_2^0 case)

If A is Δ_2^0 and not c.e., then there is an enumeration operator Φ such that $\emptyset <_e \Phi(A) <_e A$.

Proof.

Build Φ and $X = \Phi(A)$ using finite injury. Requirements:

$$\mathcal{A}_e : X \neq W_e \quad \text{and} \quad \mathcal{B}_e : \Psi_e(X) \neq A$$

Strategy for $\mathcal{B}_e$: find $a \in A \Delta \Psi_e(X)$, by making X c.e. until a is found.

Strategy for $\mathcal{A}_e$: build a reduction Ψ so that $\Psi(X) = \Psi(W_e) = A$.



Table of contents

- 1 Enumeration reducibility
- 2 Downwards density in $\mathcal{D}_e$
- 3 Computability in $L_{\omega_1^{CK}}$
- 4 Hyperenumeration reducibility
- 5 Downwards density in $\mathcal{D}_{he}$
- 6 Other reducibilities

Admissible sets

The usual definition of a Π_1^1 set of natural numbers is a set of the form $m \in X \iff \forall f \in \omega^\omega \exists n [R(f, n, m)]$ where R is a computable relation. However admissibility gives us another definition in terms of $L_{\omega_1^{CK}}$ that is useful.

Definition

A set M is *admissible* if it is transitive, closed under union, pairing and Cartesian product as well as satisfying the following two properties:

Δ_1 -comprehension: for every Δ_1 definable class $A \subseteq M$ and set $a \in M$ we have $A \cap a \in M$.

Σ_1 -collection: for every Σ_1 definable class relation $R \subseteq M^2$ and set $a \in M$ such that $a \subseteq \text{dom}(R)$ there is $b \in M$ such that $a \subseteq R^{-1}[b]$.

- The smallest admissible set is HF the collection of hereditarily finite sets. Looking at the Δ_1 and Σ_1 subsets of HF is one notion of computability. We have that the Δ_1 subsets of HF are computable sets and the Σ_1 subsets of HF are c.e. sets.
- We generalize this to an arbitrary admissible set M by calling a set, $A \subseteq M$, *M -computable*, if it is a Δ_1 subset of M , and *M -c.e.* if it is a Σ_1 subset of M .
- The smallest admissible set containing ω as an element is $L_{\omega_1^{CK}}$. We have that the $L_{\omega_1^{CK}}$ -c.e. subsets of ω are precisely the Π_1^1 sets.

Table of contents

- 1 Enumeration reducibility
- 2 Downwards density in $\mathcal{D}_e$
- 3 Computability in $L_{\omega_1^{CK}}$
- 4 Hyperenumeration reducibility**
- 5 Downwards density in $\mathcal{D}_{he}$
- 6 Other reducibilities

Definition (Sanchis 1978)

We say that A is *hyperenumeration reducible* to B (written $A \leq_{he} B$), if there is a c.e. set W such that

$$n \in A \iff \forall f \in \omega^\omega \exists u \in \omega, x \prec f[\langle n, x, u \rangle \in W \wedge D_u \subseteq B]$$

- Like with enumeration reducibility, this is a pre-order and the equivalence classes give us the hyperenumeration degrees, $\mathcal{D}_{he}$.
- From an effective listing of c.e. sets $(W_e)_{e \in \omega}$, we obtain an effective listing of hyperenumeration operators $(\Gamma_e)_{e \in \omega}$.

Relating $\leq_h$ and $\leq_{he}$

Hyperarithmetical reducibility, defined by $A \leq_h B \iff A \in \Delta_1^1(B)$, has a close relationship with hyperenumeration reducibility:

Theorem (Sanchis)

- For any sets A, B we have that A is $\Pi_1^1(B)$ if and only if $A \leq_{he} B \oplus \overline{B}$.
- The map $A \mapsto A \oplus \overline{A}$ induces an embedding of $\mathcal{D}_h$ in $\mathcal{D}_{he}$.
- There is a degree that is not in the range of this embedding, i.e. not all he -degrees are hypertotal.

Theorem (J-G)

There is a set A such that $A <_{he} \overline{A}$ but
 $\{X : A \leq_{he} X \oplus \overline{X}\} = \{X : \overline{A} \leq_{he} X \oplus \overline{X}\}.$

So Selman's theorem fails in this context.

Sanchis proved an interesting result about the relationship between enumeration reducibility and hyperenumeration reducibility.

Theorem (Sanchis)

If $A \leq_e B$ then $A \leq_{he} B$ and $\bar{A} \leq_{he} \bar{B}$.

- This means that if $f : \omega \rightarrow \omega$ is an enumeration of A , then $A \oplus \bar{A} \leq_{he} f$.
- So the hypertotal degrees above A are not necessarily limited to the degrees containing enumerations of A .

Table of contents

- 1 Enumeration reducibility
- 2 Downwards density in $\mathcal{D}_e$
- 3 Computability in $L_{\omega_1^{CK}}$
- 4 Hyperenumeration reducibility
- 5 Downwards density in $\mathcal{D}_{he}$
- 6 Other reducibilities

A hyper Gutteridge operator

Theorem (J-G)

There is an hyperenumeration operator, Λ , such that for any set X we have that:

- *If $X \leq_{he} \Lambda(X)$ then X is Π_1^1 .*
- *$X \leq \Lambda(X) \oplus \overline{\mathcal{O}}$.*

Proof.

Build $L_{\omega_1^{CK}}$ -c.e. set $B \subseteq \omega$ defined by $B = \bigoplus_k \{x < n_k\}$.

$\Lambda(X) = B \cup \{\langle k, n_k \rangle : k \in X\}$.

$k \in X \iff \exists n[\langle k, n \rangle \in \overline{B} \cap \Lambda(X)]$, so $X \leq \Lambda(X) \oplus \overline{\mathcal{O}}$.

For all $m > e$ and finite $D \subseteq m$ we have the following requirement:

$$\mathcal{R}_{e,m,D} : m \in \Gamma_e(\Lambda(D)) \iff m \in \Gamma_e(\Lambda(D \cup \{n : n \geq m\}))$$



The issue with injury arguments in this context

Proposition

For all enumeration operators, Ψ , we have that

$$\Psi(X) = \bigcup_{D \subseteq_{\text{fin}} X} \Psi(D)$$

In priority arguments this is used when we build $X = \lim_s X_s$ and claim that if $n \in \Psi(X)$ then there is s such that $n \in \Psi(X_s)$.

Proposition

There is a hyperenumeration operator, Γ , and set X such that

$$\Gamma(X) \not\supseteq \bigcup_{\{H \subseteq X : H \leq_{\text{he}} \emptyset\}} \Gamma(H)$$

A tool to ensure non-hyper enumeration

Proposition

If A is $L_{\omega_1^{CK}}$ -c.e. then $\Gamma_e(A) = \bigcup_{H \subseteq_{\text{hyp}} A} \Gamma(H)$.

In the construction of Λ we ensure that for $n \geq e$:

- $n \in \Gamma_e(\Lambda(A)) \iff n \in \Gamma_e(\Lambda(A \upharpoonright n))$.
- $\Lambda(A \upharpoonright n)$ is a Π_1^1 set.

So $n \in \Gamma_e(\Lambda(A)) \iff \exists H \subseteq_{\text{hyp}} A \Lambda(A \upharpoonright n)[n \in \Gamma_e(H)]$.

Theorem (J-G)

If $A = \lim_{s \rightarrow \omega_1^{CK}} A_s$ and the map $s \mapsto A_s$ is $L_{\omega_1^{CK}}$ -computable, then there is an $L_{\omega_1^{CK}}$ -c.e. operator Φ , such that $\emptyset <_{he} \Phi(A) <_{he} A$.

Tools used in the proof:

- We define $X = \Phi(A) = \{x : \exists H \in L_{\omega_1^{CK}}[(x, H) \in \Phi \wedge H \subseteq A]\}$.
- We ensure that $X \subseteq \omega_1^{CK} \times \omega_1^{CK}$ and for each α , $X^{[\alpha]}$ is an ordinal. (converted to a subset of $\omega \times \omega$ using a Harrison order).
- The finite injury argument becomes an infinite injury argument using a tree of strategies. The idea is that each node σ has a guess $\hat{A}_\sigma$ to A .
- σ only adds axioms of the form $(\langle \alpha, \beta \rangle, \hat{A}_\sigma)$ so nodes to the right of the true path do not effect $\Phi(A)$.
- Along the true path p we have $A = \bigcup_{\sigma \prec p} \hat{A}_\sigma$.

Table of contents

- 1 Enumeration reducibility
- 2 Downwards density in $\mathcal{D}_e$
- 3 Computability in $L_{\omega_1^{CK}}$
- 4 Hyperenumeration reducibility
- 5 Downwards density in $\mathcal{D}_{he}$
- 6 Other reducibilities

The failure of Selman's theorem

Recall the following results about $\leq_e$ and $\leq_{he}$:

Theorem (Selman's theorem)

For any sets A, B we have

$$A \leq_e B \iff \{X : A \leq_e X \oplus \overline{X}\} \supseteq \{X : B \leq_e X \oplus \overline{X}\}$$

Theorem (J-G)

*There is a set A such that $A <_{he} \overline{A}$ but
 $\{X : A \leq_{he} X \oplus \overline{X}\} = \{X : \overline{A} \leq_{he} X \oplus \overline{X}\}$.*

This suggests that perhaps $\leq_{he}$ is not the “correct” notion of hyperenumeration reducibility.

Question

Are there other higher computability analogues of enumeration reducibility?

Reducibilities weaker than $\leq_{he}$

Definition

- We say that A is *relatively Π_1^1 in B* , $A \leq_{\Pi_1^1} B$, if whenever B is Π_1^1 in X we have that A is Π_1^1 in X .
- We say that A is *uniformly relatively Π_1^1 in B* , $A \leq_{u\Pi_1^1} B$, if there is a computable f such that if $B = \Gamma_e(X \oplus \bar{X})$ then $A = \Gamma_{f(e)}(X \oplus \bar{X})$.

Proposition

- If $A \leq_{he} B \implies A \leq_{u\Pi_1^1} B$ and $A \leq_{u\Pi_1^1} B \implies A \leq_{\Pi_1^1} B$.
- There is a set A such that $A <_{he} \bar{A}$ and $A \equiv_{u\Pi_1^1} \bar{A}$.

Question

Does $A \leq_{\Pi_1^1} B$ imply $A \leq_{u\Pi_1^1} B$?

Reducibilities stronger than $\leq_{he}$

Definition

- We say that A is *continuously higher enumeration reducible* to B , $A \leq_{che} B$ if there is a Π_1^1 set W such that $n \in A \iff \exists u[\langle n, u \rangle \in W \wedge D_u \subseteq B]$.
- We say that A is ω_1^{CK} -*enumeration reducible* to B , $A \leq_{\omega_1^{CK}} B$, if there is an $L_{\omega_1^{CK}}$ -c.e. set W such that $n \in A \iff \exists H[(n, H) \in W \wedge H \subseteq B]$

Theorem (J-G)

$A \leq_{che} B \implies A \leq_{\omega_1^{CK}} B$ and $A \leq_{\omega_1^{CK}} B \implies A \leq_{he} B$, and these implications are strict.

Theorem (J-G)

The che-degrees are isomorphic to the e-degrees above $\mathcal{O}$.

Question

What is the “correct” notion of hyperenumeration reducibility:

$\leq_{he}$, $\leq_{u\Pi_1^1}$, $\leq_{\Pi_1^1}$, something else?

Question

Are the hypertotal degrees definable in $\mathcal{D}_{he}$?

How complex is the theory of $\mathcal{D}_{he}$?

Thank you

Thank You